Epigenetic drugs: reducing the transgenerational effect of maternal dietary exposures on offspring mammary cancer risk

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ABSTRACT

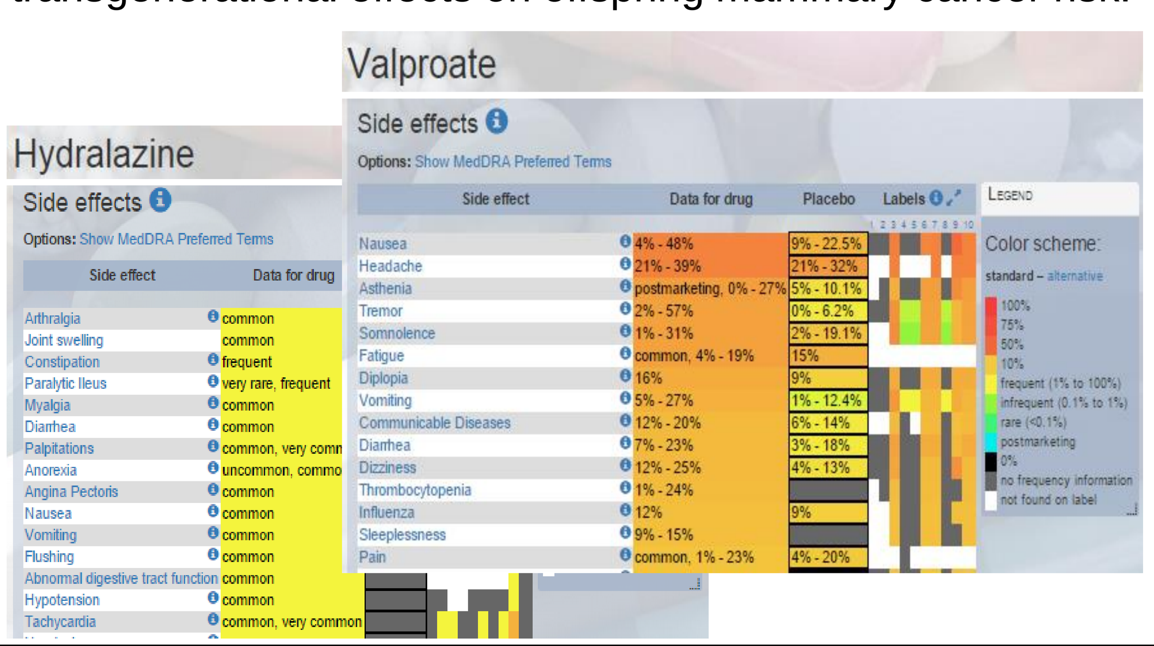
Maternal dietary exposures to high levels of polyunsaturated fatty acids (PUFA) adversely change the epigenome of subsequent generations of offspring. Previous studies in our laboratory discovered that due to their in utero high-fat exposure, offspring have higher risks and more aggressive cases of mammary cancer, as the diet increases both F1 and F3 generations' tumor incidence (F1: $p<0.0382$), burden (F1: $p<0.0001$, F3: $p<0.0001$) and malignancy; we hypothesized that the observed results were regulated by epigenetic mechanisms. The apparent permanence of the transgenerational effects of the diet, however, can be reversed through the use of valproic acid, a histone deacetylase (HDAC) (1.2 g/kg), and hydrazine a DNA methyltransferase (DNMT) (5 mg/kg) inhibitors, given via drinking water. F1 and F3 lineage that were in utero-exposed to F0 maternal high-fat diet were treated with these inhibitors. While we support earlier studies that presented a worse mammary cancer prognosis among control-diet mice following the use of epigenetic drugs, our findings reveal that offspring of high-fat female mice significantly benefited from daily epigenetic drug treatments that began before mammary tumorigenesis (F1 incidence: $p<0.1024$, burden: $p<0.0082$). Such encouraging prognoses may ultimately lead to the translation of the drug combination into clinical cancer prevention

BACKGROUND

Breast cancer is the second leading cause of cancer death in American women, and about 1 in 8 will develop invasive breast cancer during their lifetime[1]. While effective cancer therapies are on the rise, currently there are endeavors to progress cancer prevention. Environmental and lifestyle factors, such as everyday exposures to chemicals, physical activity, and diet, have been discovered to strongly correlate with cancer incidence and aggression.

It was previously studied that maternal dietary exposures influence the risk of breast cancer in subsequent generations. In particular, there is significant evidence that a maternal PUFA-rich diet, which henceforth will be identified as generation F0, *even* increases breast cancer risk for F3 offspring[2], alluding to a certain permanence of the diet's effects. These findings thus concluded that the high-fat diet affected the offspring epigenome through changes in methylation and acetylation patterns. Considering these epigenetic changes, we investigated possible measures to alleviate the adverse effects.

Epigenetic therapy has been recently recognized as an effective method of treatment for various cancers. Combinations of epigenetic-targeted drugs, such as the HDAC inhibitor Valproic acid and DNMT inhibitor Hydralazine[3], are traditionally used as a final treatment option for cancer patients[4]. However, our study demonstrates that HDAC and DNMT inhibitors may assume roles beyond curative measures. In the form of daily preventive medication, epigenetic drugs may be the means to reverse maternal transgenerational effects on offspring mammary cancer risk.



Most common side effects of HDAC inhibitor valproic acid (valproate) and DNMT inhibitor hydralazine.

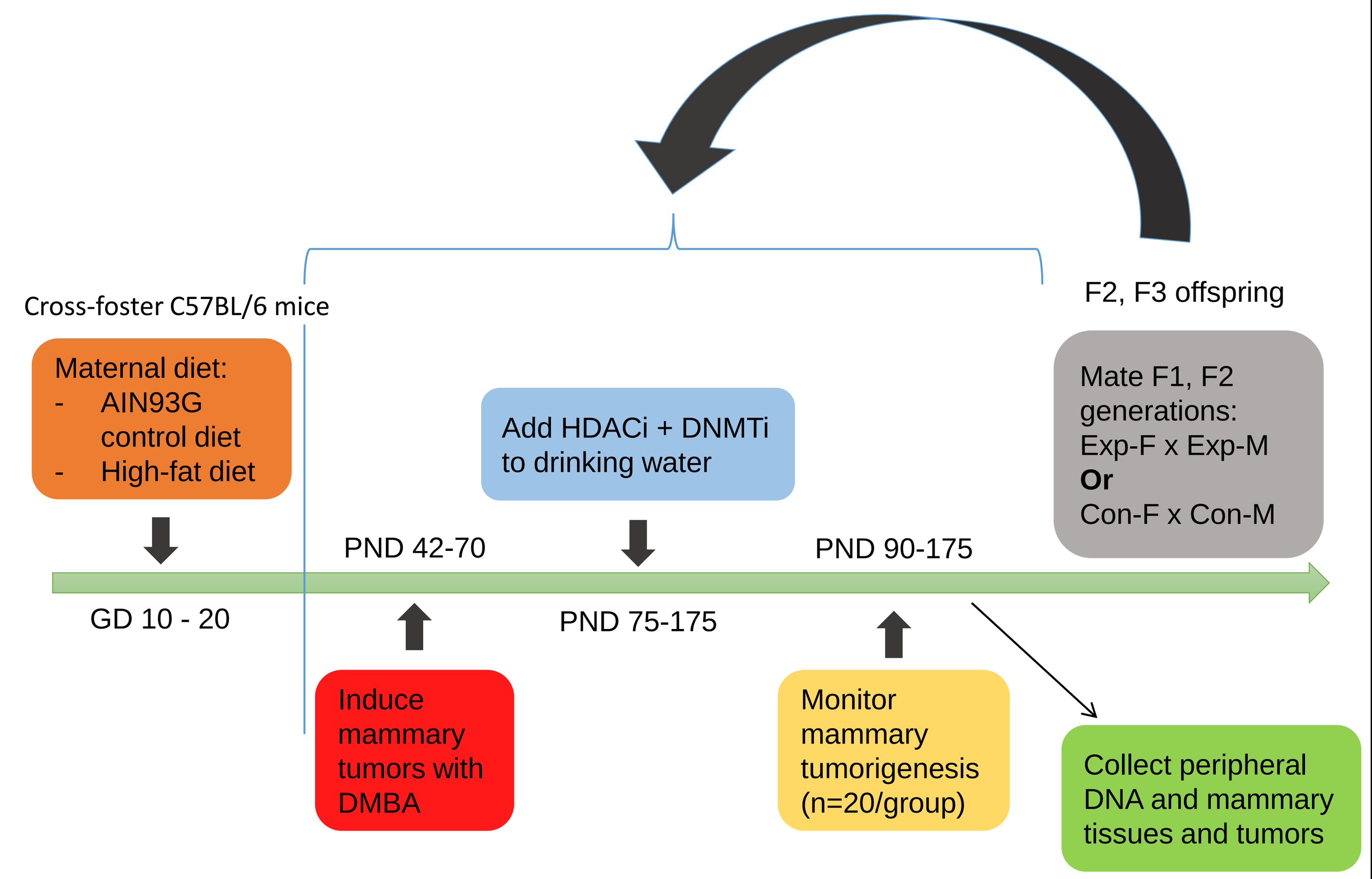
METHODOLOGY

Maternal Exposure to High-Fat Diet

On gestation day (GD) 10, while one F0 group of pregnant C57BL/6NTac mice continued to be fed the control diet (6% corn oil), the F0 high-fat (HF) group was fed an isocaloric high n-6 polyunsaturated fat diet (18% corn oil) until GD 20. The HF lineage were only exposed to a dietary change during this time period.

Epigenetic Drug Treatment

On post-natal day (PND) 75, both control and HF mice were given valproic acid and hydralazine at a low dosage in their drinking water. They received constant exposure to the drug solution for about twenty weeks or until euthanasia is necessary. Each mouse consumed 1.2 g/kg of valproic acid and 5 mg/kg of hydralazine per day.



RESULTS

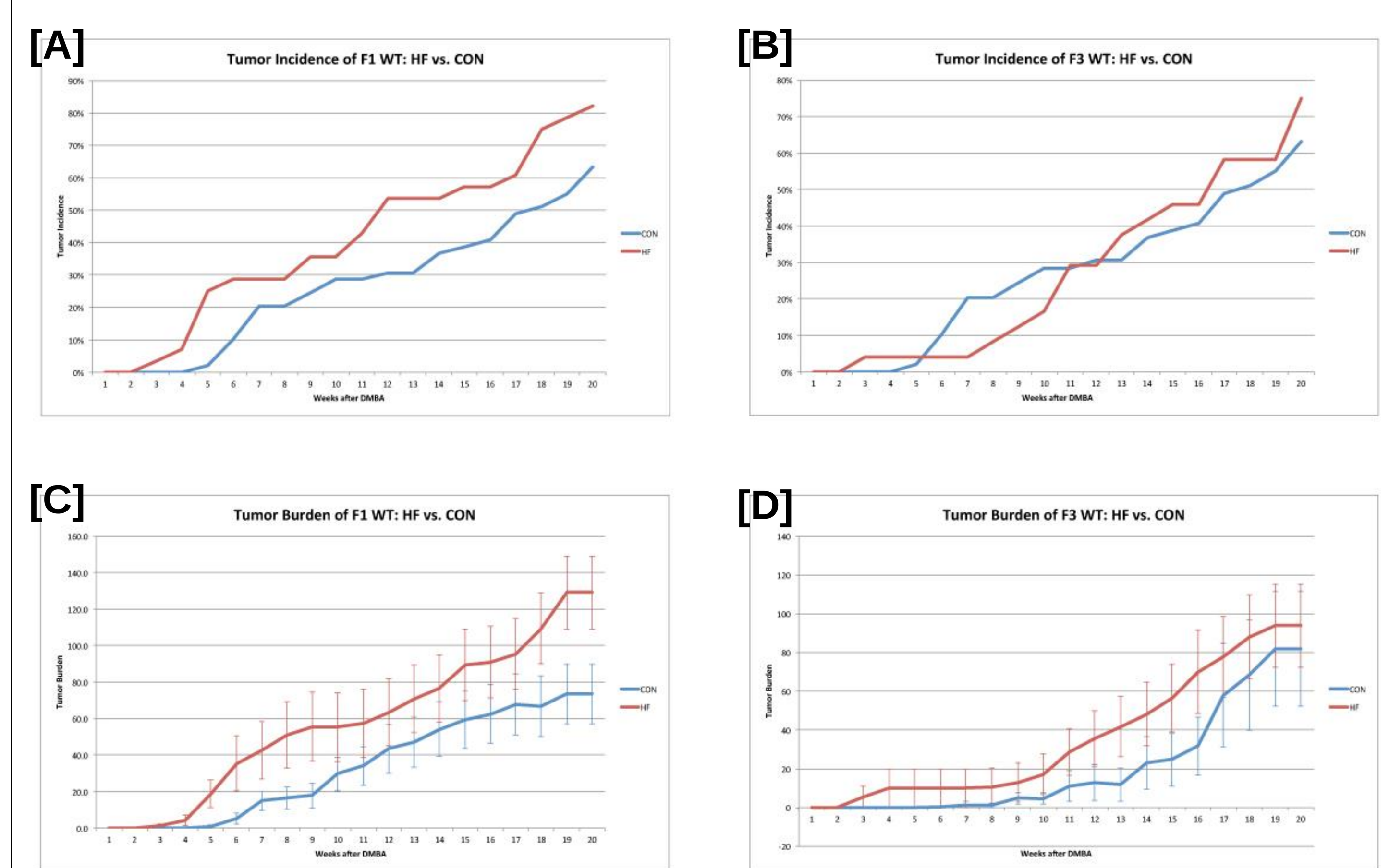


Figure 1. Maternal HF exposure increases tumor incidence and burden up to F3 generation. [A] F1 HF *in utero*-exposed female offspring had significantly higher mammary tumor incidence than F1 CON female offspring (log-rank test: $p<0.0382$).
[B] F3 groups began to have a significant difference in tumor incidence from week 12 until the end of the study period.
[C],[D] F1 and F3 HF mice exhibited significantly greater tumor burden than control mice (paired t-test: F1 $p<0.0001$, F3 $p<0.0001$).

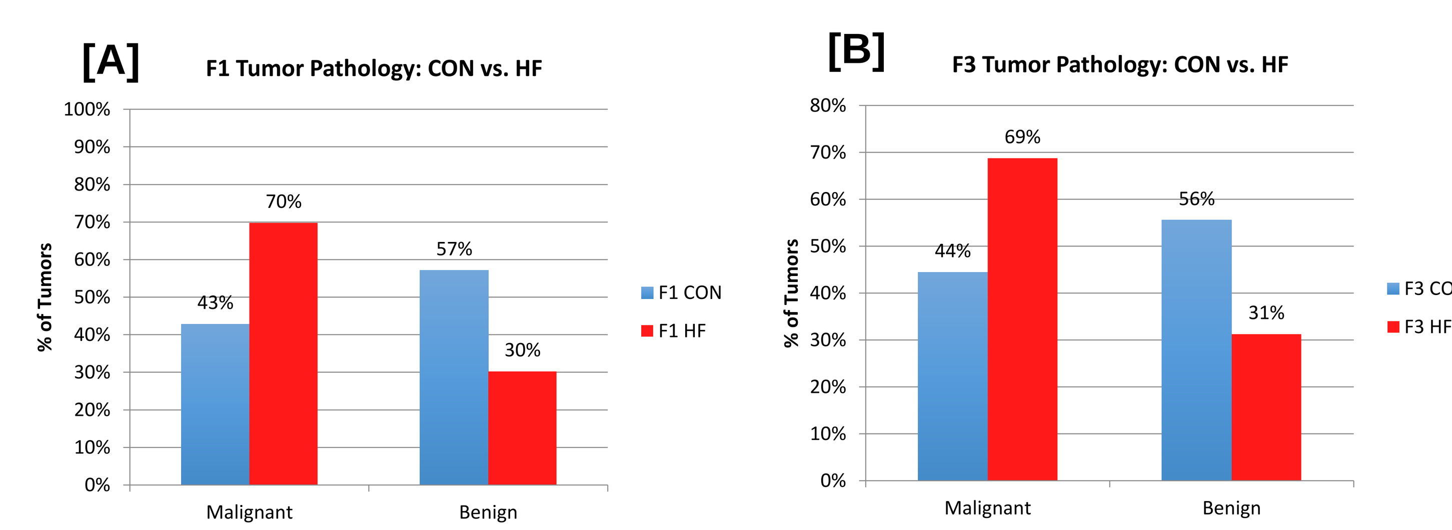


Figure 2. Maternal HF dietary exposure increases tumor malignancy for F1 – F3 offspring. [A] Using stained histopathology slides, 70% of tumors collected from F1 HF *in utero*-exposed offspring were identified as malignant, in comparison to 43% of CON offspring.
[B] From F3 HF offspring, 69% of tumors collected are considered as malignant, in comparison to 44% of CON offspring.

RESULTS (cont.)

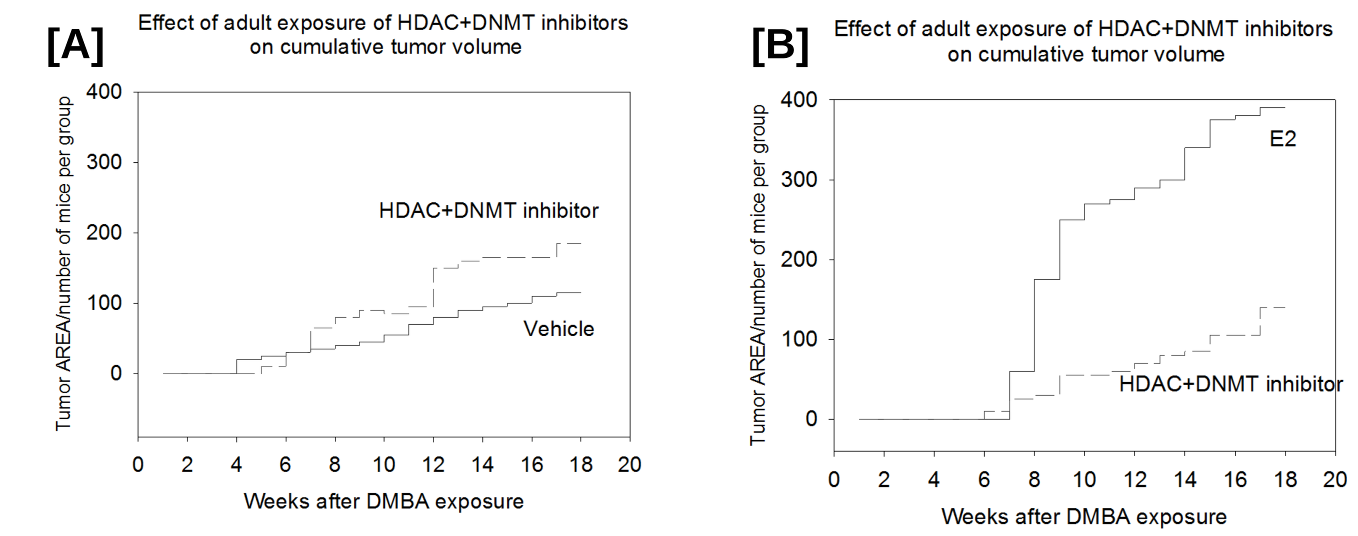


Figure 3. [A] Earlier studies indicated that in control mice, HDACi + DNMTi drugs do not inhibit tumor growth when compared to the vehicle, which was the mice group on the control diet. [B] Yet tumors of *in utero* estradiol-exposed mice significantly reduced in size after epigenetic drug treatment.

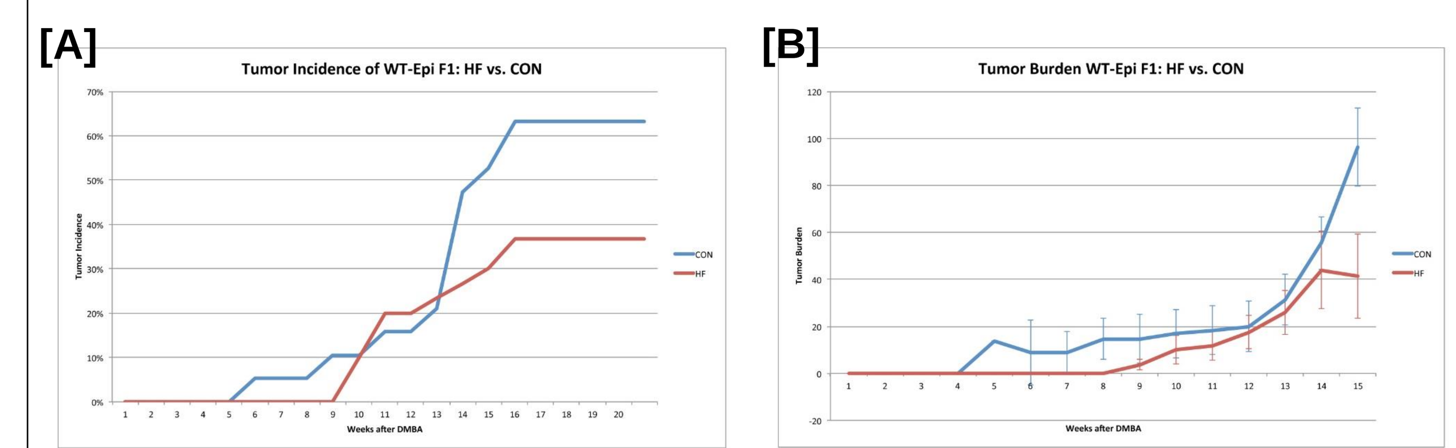


Figure 4. HDAC and DNMT inhibitors curb tumor incidence and burden for HF *in utero*-exposed offspring. For F1 HF offspring, [A] tumor incidence decreases ($p<0.1024$) and [B] tumor burden also decreases ($p<0.0082$).

DISCUSSION / CONCLUSION

- Our findings support the hypothesis that *in utero* exposure to a high-fat diet during critical periods of embryonic development can increase breast cancer risk transgenerationally, as tumors with a greater level of aggression proliferated in numbers.
- In pre-clinical rodent models, mammary cancer risk in *in utero*-exposed offspring decreases to normal levels when the mice consume epigenetic drugs as preventive medication. Valproic acid and hydralazine effectively reversed this transgenerational change.
- Because the two drugs in solution are low in toxicity, are FDA-approved—thus widely accessible—and cause a few side effects, they have a promising future in breast cancer prevention.
- For the successful transition of epigenetic drugs into clinical preventive practice, we must identify and characterize biomarkers that would appropriately indicate at-risk individuals. Following these studies can proper, timely methods of breast cancer prevention be administered.

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